

Distributed Deep Learning on GPU-based Clusters

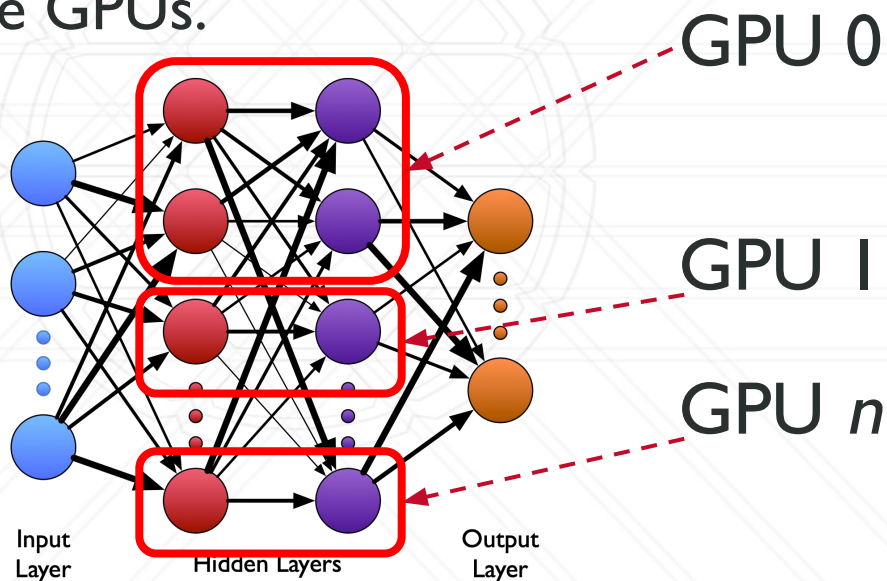
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Limitations of data parallelism

- DDP – Supports models of limited size
- FSDP – Large communication overheads, specially for small batch size tasks like IFT

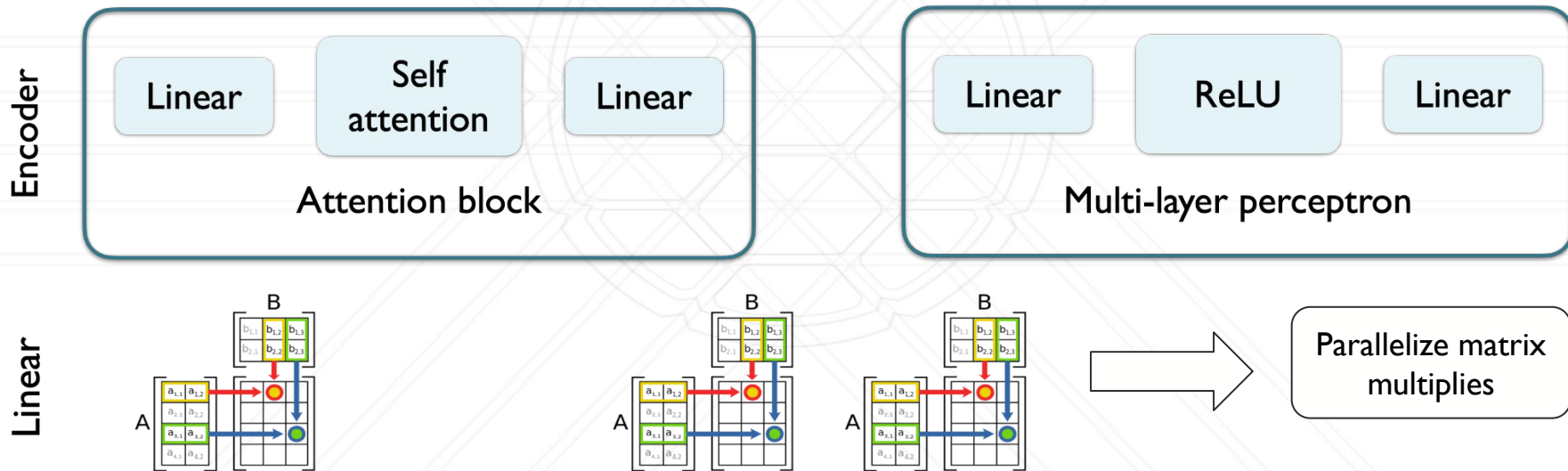
Tensor parallelism

- Divide parameters and compute of every layer of a neural network on multiple GPUs.



Tensor parallelism

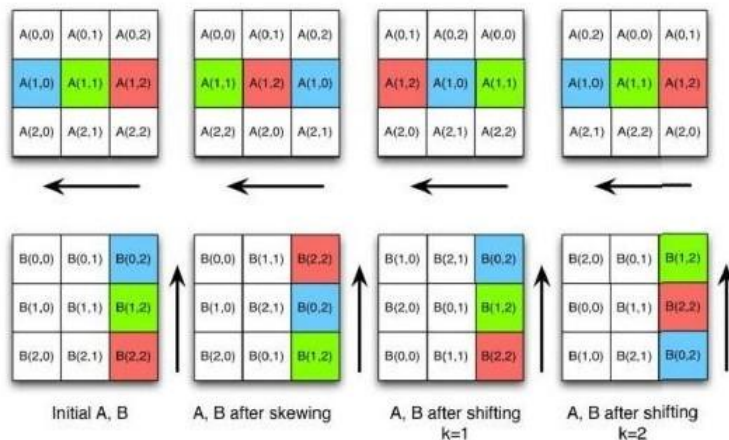
- Divide parameters and compute of every layer of a neural network on multiple GPUs.



Intra-layer parallelism

- Divide parameters and compute of every layer of a neural network on multiple GPUs.
- Two kinds of layers
 - ReLU and Layernorm – apply same function to each element of the input tensor
 - Fully Connected/Convolution – matrix multiplications

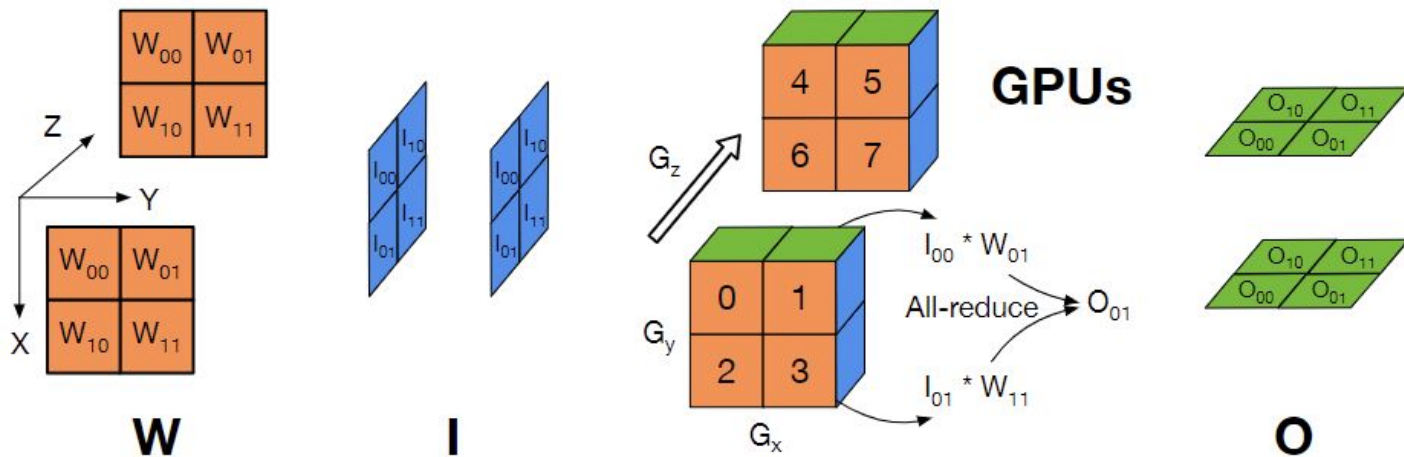
Parallelizing a Matrix Multiplication



$$C(1,2) = A(1,0) * B(0,2) + A(1,1) * B(1,2) + A(1,2) * B(2,2)$$

Cannon's Parallel Matrix Multiplication Algorithm

AxoNN's 3D Tensor Parallelism



Parallelizing a matrix multiplication ($I \cdot W = O$) using AxoNN on 8 GPUs

Creating an AxoNN Lightning Strategy

```
from axonn.lightning import AxonnStrategy
```

```
pl_strategy = AxonnStrategy(
```

```
    G_intra_r=..
```

```
    G_intra_c=..
```

```
    G_intra_d=..
```

```
    overlap_communication=True,
```

```
)
```

3D tensor parallel grid
dimensions

Running the code (Tensor/Intra-Layer)

- Code – `train.py`

```
CONFIG_FILE=configs/axonn.json sbatch train_multi.sh
```

Let's try different AxoNN configurations

- In config/axonn.json - tp_dimensions is [2, 2, 1].
- Now change it to [4, 1, 1] and rerun your code.
- Does it become faster?

How to set the parallel configuration?

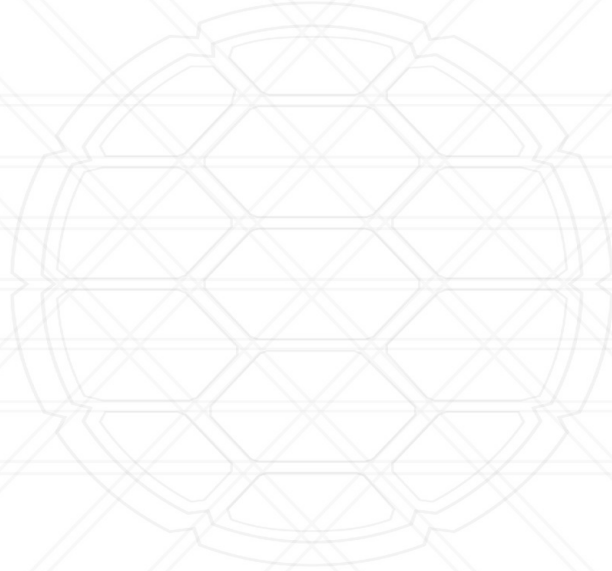
- Small batch-sizes - Use x and/or y
 - Example - finetuning and inference
- Large batch sizes - Use z (+data parallelism)
 - Example - pretraining

Alternative way to use AxoNN

```
from axonn.intra_layer import auto_parallelize
with auto_parallelize():
    net = FC_Net(args.num_layers, args.image_size**2, args.hidden_size, 10).cuda()
```

Zero code changes required in your model definition!

`fabric.init_module()` calls AxoNN's auto-parallelize function



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